



Product Review System

PROJECT NLP | BUSINESS CASE: AUTOMATED
CUSTOMER REVIEWS

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Project Introduction

Project Objective:

Develop a product review system powered by NLP models that aggregate customer feedback from different sources.

Approach:

- Classifying reviews
- Clustering product categories
- Using generative AI to summarize reviews into recommendation articles.

Dataset:



Amazon Product Reviews

- This is a large-scale Amazon Reviews dataset collected in 2023. It contains 48.19 million items, and 571.54 million reviews from 54.51 million users.

Basic statistics

Ratings: 571.54 million

Users: 54.51 million

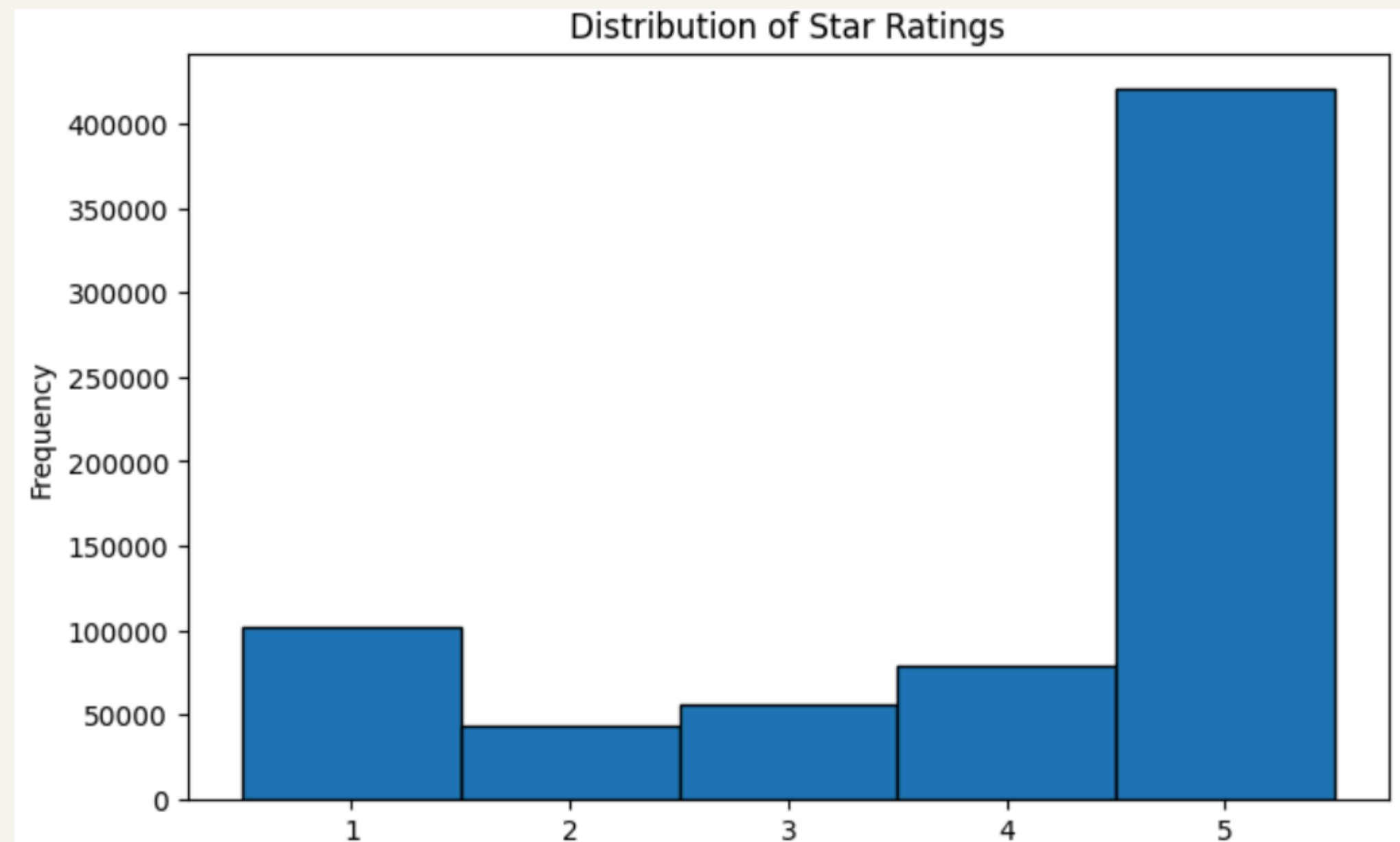
Items: 48.19 million

Timespan: May 1996 - September 2023

Dataset Observations



1. Rating Distribution is Skewed



Dataset Observations



- 2. Average review length is moderate (~50–100 words)
- 3. Product Diversity
- 4. Text Content is Mostly Clean



Review Classification

- Classify customer reviews into three sentiment categories: Positive, Neutral, and Negative based on their textual content.
- Used a pre-trained transformer model (DistilBERT) from Hugging Face.

Preprocessing steps:



1

Load and split the data into train and test sets

2

Sentiment Label Mapping

- 1–2 stars → "negative" (label = 0)
- 3 stars → "neutral" (label = 1)
- 4–5 stars → "positive" (label = 2)

3

Tokenization

Used DistilBertTokenizerFast to tokenize the review text column

4

Class Imbalance Handling

- Computed class weights using `sklearn.utils.class_weight` to handle imbalance

5

Set Format for PyTorch

- Specified the required input columns: `input_ids`, `attention_mask`, and `label`
- Used `.set_format(type="torch", columns=[...])` to prepare data for the Trainer

Fine-Tuning Process



Training Arguments:

- `eval_strategy="epoch"`
- `num_train_epochs=5`
- `learning_rate=2e-5`
- `weight_decay=0.01`
- `fp16=True`

Trainer Setup:

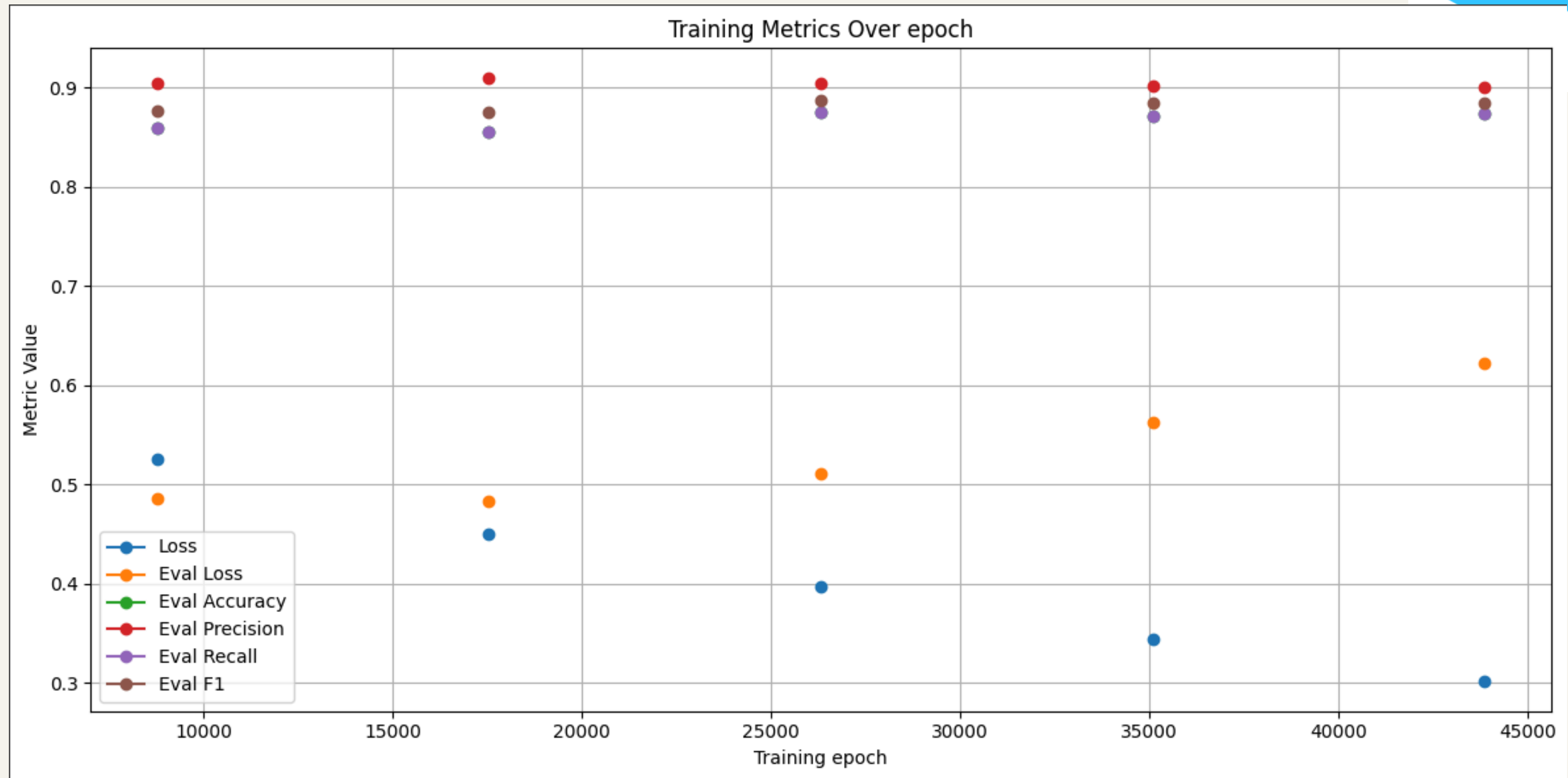
- `trainer = WeightedLossTrainer`
- `model=model,`
- `args=training_args,`
- `train_dataset=train_dataset`
- `eval_dataset=test_dataset`
- `compute_metrics`

Classification Result



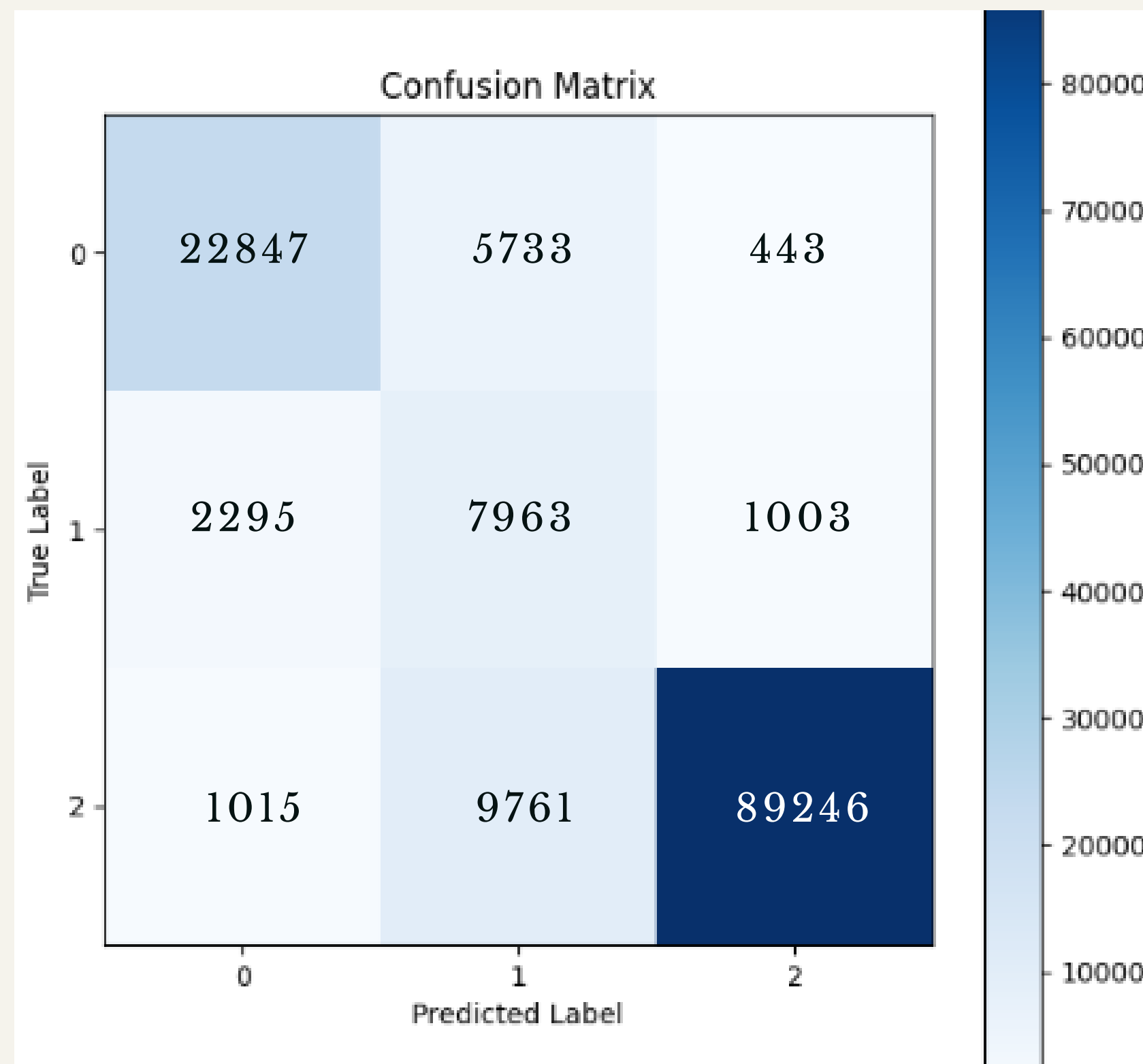
Matrices	precision	recall	f1-score	accuracy
negative	0.87	0.79	0.83	0.86
neutral	0.34	0.71	0.46	
positive	0.98	0.89	0.94	

Classification Result





Classification Result

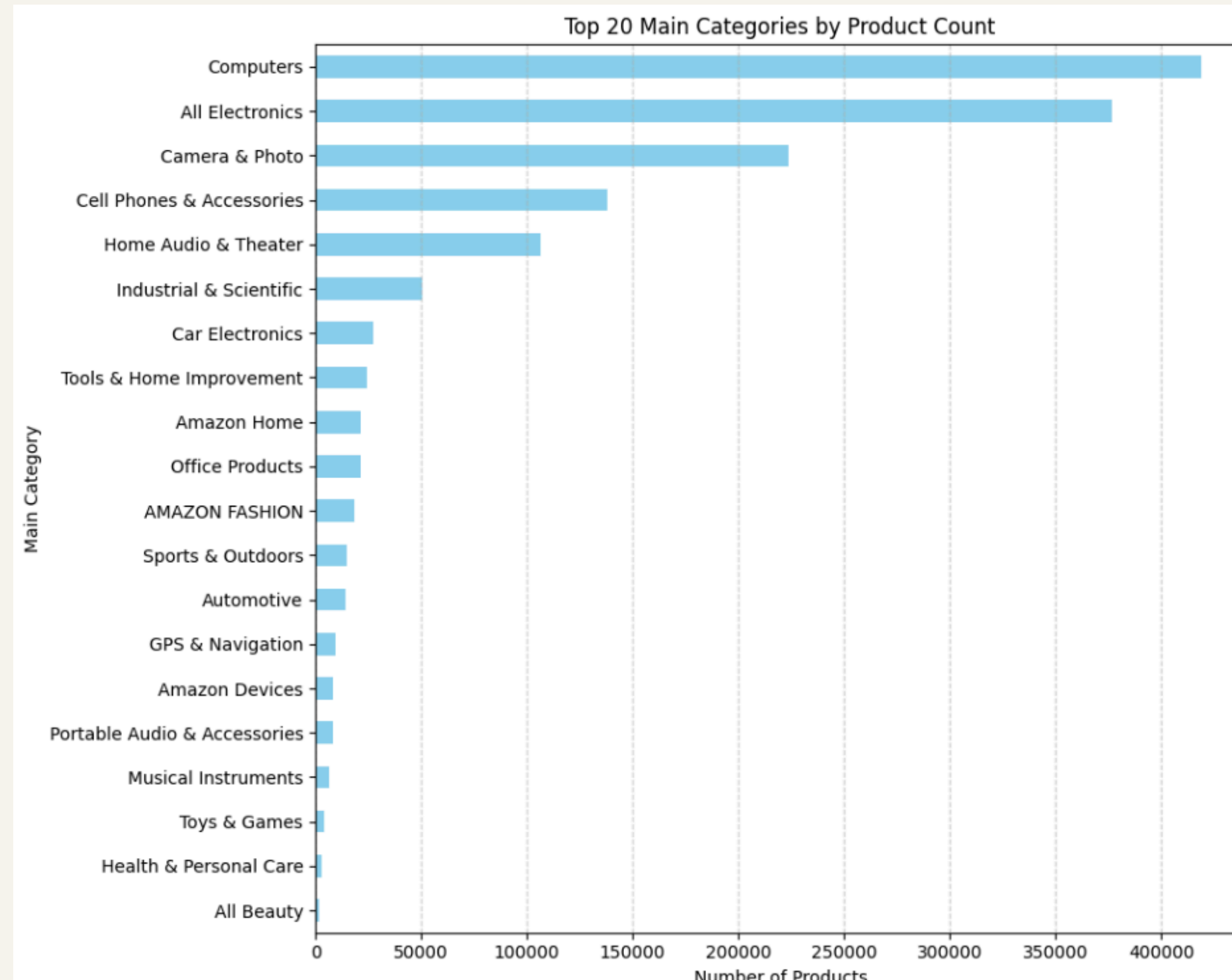




Product Category Clustering

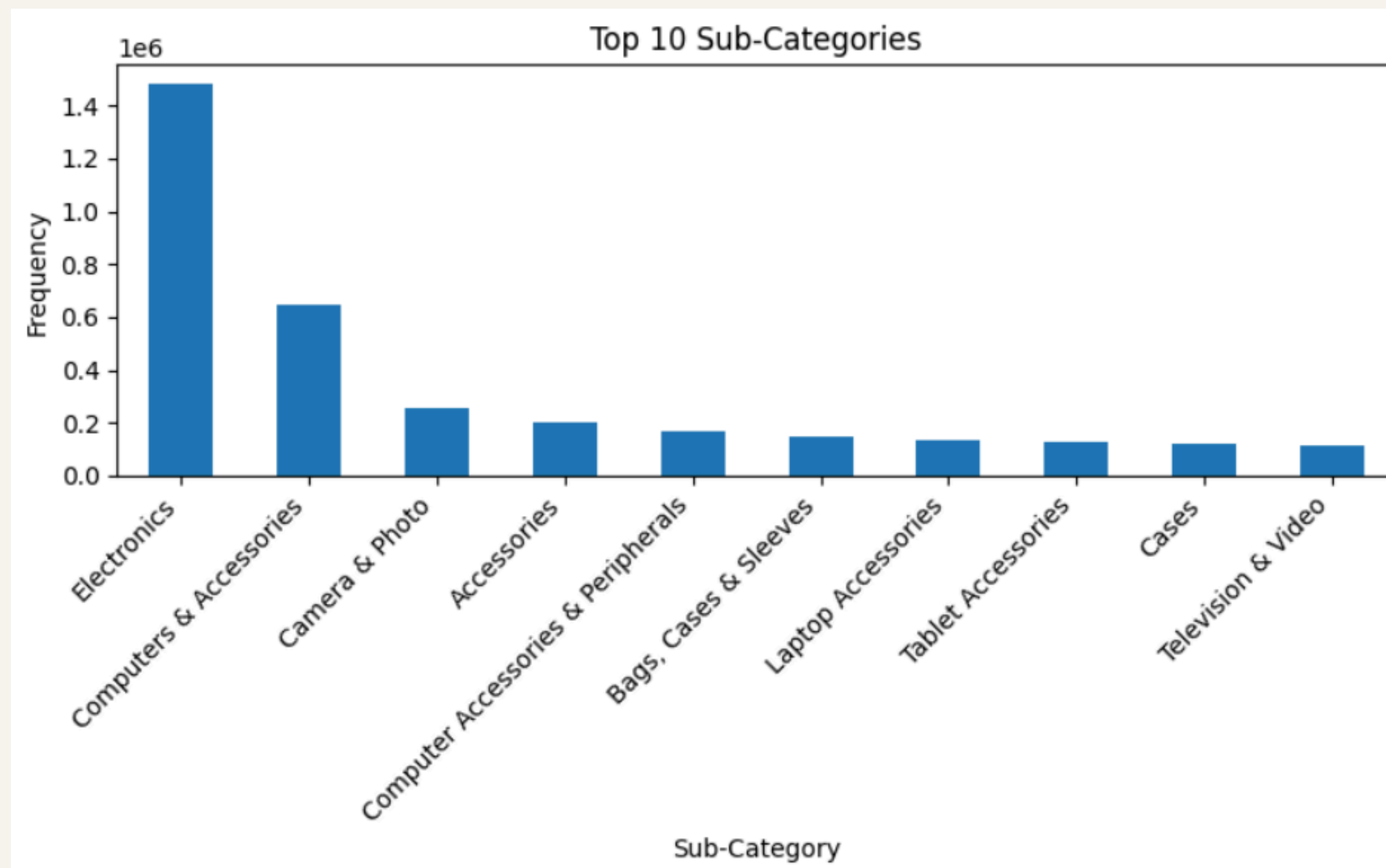
- Group products into broader meta-categories based on their titles, simplifying review analysis and recommendation generation.
- Applied K-Means clustering and tuned the number of clusters using:
 - WCSS and Elbow Method

Exploratory Data Analysis for Clustering





Exploratory Data Analysis for Clustering



Preprocessing steps:



1

Selected Columns:

- Used only parent_asin and title columns from the metadata (meta_df).

2

Text Vectorization (TF-IDF):

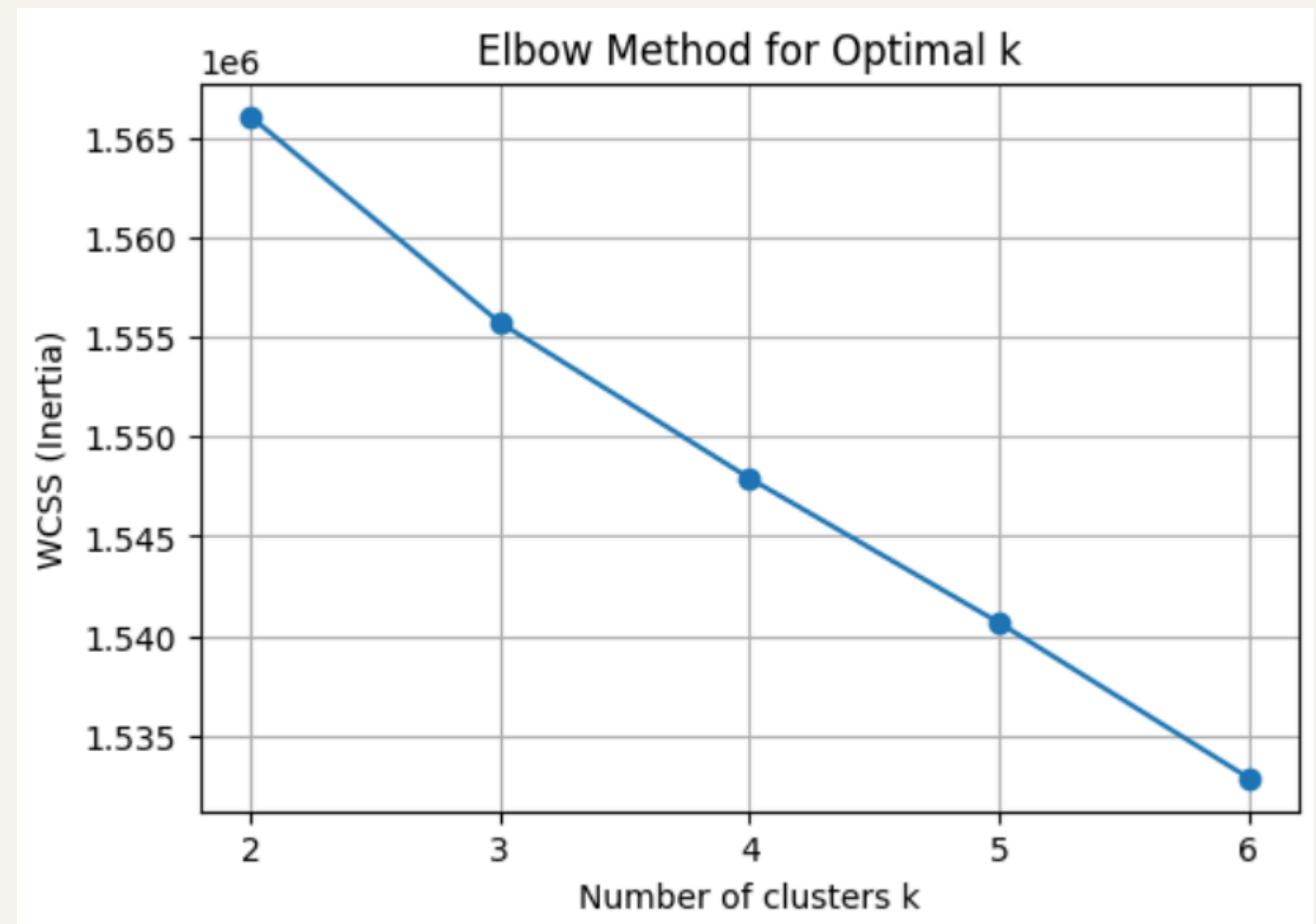
- Applied TF-IDF vectorization on the product titles.
- Limited the number of features (max_features=5000) and removed stop words (stop_words='english').

Clustering: K-mean



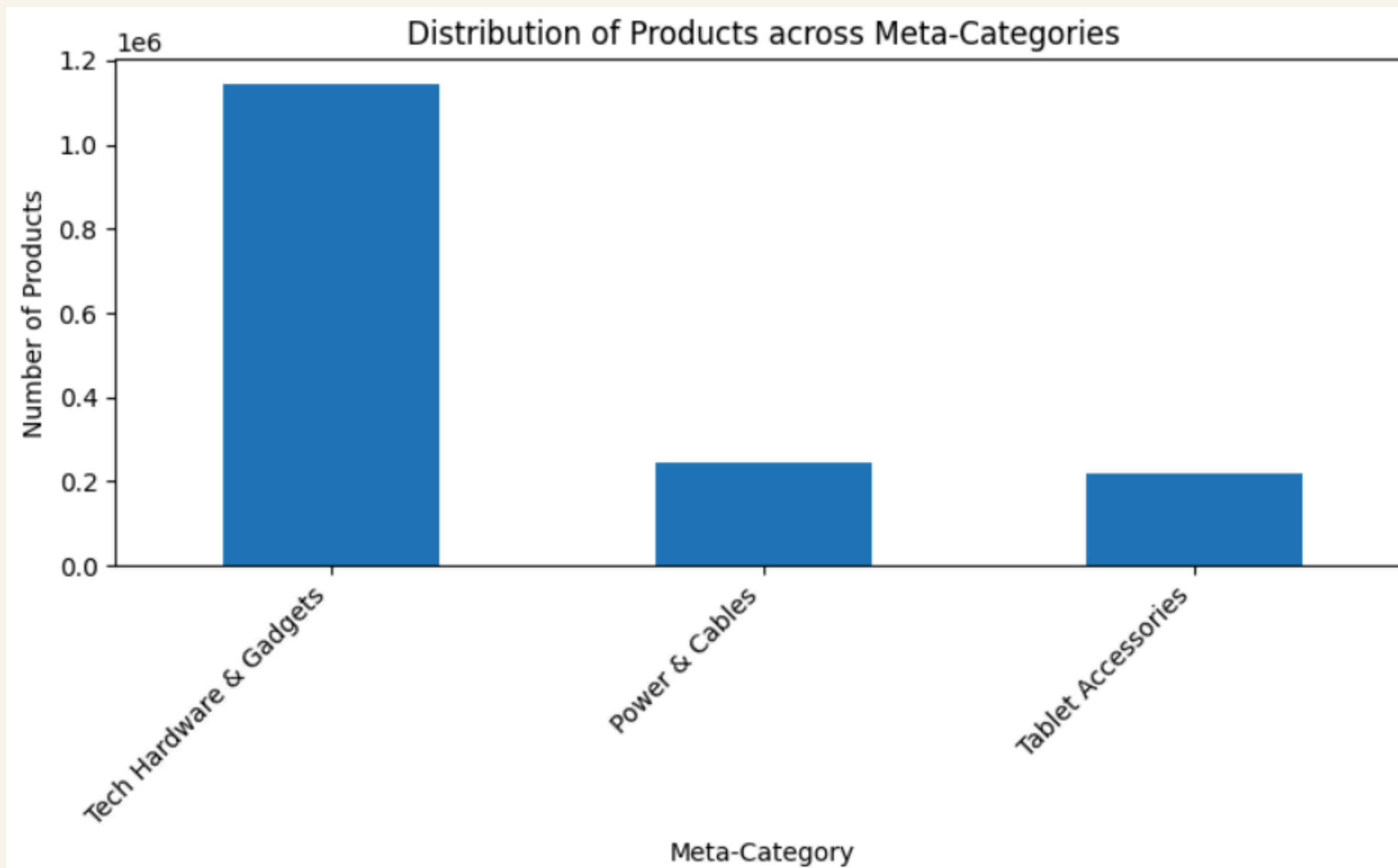
Training:

- `n_clusters = 3` based on elbow analysis
- Random seed = 42





Clustering Result





Review Summarization

- Goal: Generate blog-style summaries recommending products within each meta-category based on user reviews.
- Used gpt-3.5-turbo to generate summaries.



Dataset Preparation

- Merged review data (text, rating, parent_asin) with product metadata (title, main_category).
- Grouped reviews by main_category for category-level summarization.

OpenAI Generation Parameters

- temperature=0.7
- max_tokens=600
- top_p=0.9



Project Challenges

- Class imbalance in sentiment classification
- Determining the optimal number of clusters (KMeans)
- Merging and cleaning large-scale multi-source datasets



DEMO

REL TEAM



Thank you for listening

ANY QUESTIONS?



Project Repo:

<https://github.com/Alrumaysaa-Alghamdi/Project-NLP-Business-Case-Automated-Customer-Reviews>